IP addresses

the street address of every device

Watch the 9-min video

In one line: an IP address is a device's address on a network, with a network part (which network, like the building) and a device part (which device, like the flat number); IPv4 is 32 bits, about 4.3 billion addresses, so homes use private addresses behind one public address, and DHCP keeps them unique.

WORDS YOU'LL MEET TODAY

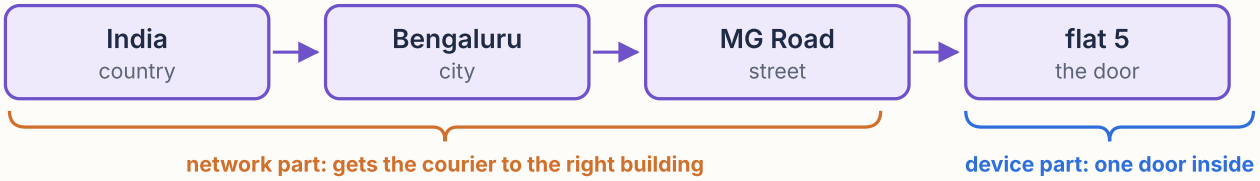
- **IP address (internet protocol address)**: a device's address on a network
- **IPv4**: the classic kind: four numbers 0 to 255, 32 bits
- **IPv6**: the newer kind: 128 bits, far more addresses
- **Public / private address**: reachable from the internet / only inside a network
- **Router**: the box that joins networks and passes packets on
- **DHCP (dynamic host configuration protocol)**: lends each device a free address
- **Bit / byte**: a bit is one 0 or 1; 8 bits make a byte
- **Localhost (127.0.0.1)**: always means this same device

1 The question: how does a message find one phone?

You type a friend's name in WhatsApp and press send. Out there are about two billion other phones, yet your message reaches exactly one: your friend's. A courier does the same thing every day: out of a million flats, he finds the right door, because every flat has an address. So what is a computer's address, what does it look like, and what happens when two devices share one?

2 An everyday picture first: the courier

A courier has a parcel for **flat 5, MG Road, Bengaluru, India**. He goes country, then city, then street; only then does he look for the flat. There are two parts: India, Bengaluru, MG Road gets him to the right building, and flat 5 finds one door inside it. And no two flats in a building share a number, or he wouldn't know where to knock.



No two flats in one building share a number; if two doors said 5, the courier would not know where to knock.

Figure 1. The courier's address: country, city and street get him to the right building; the flat number finds one door inside it.

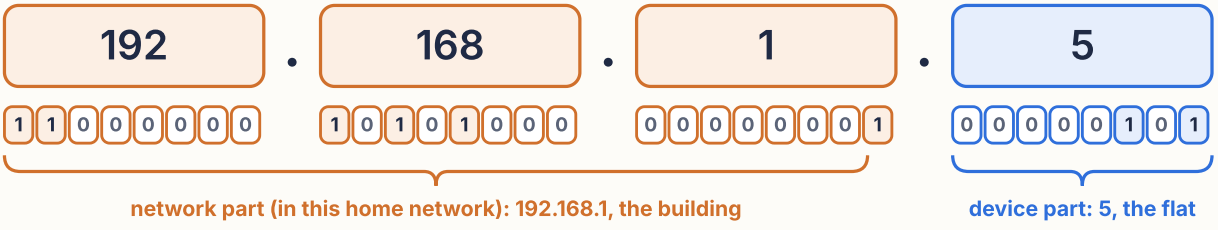
3 Now the real thing

The courier	In a network
The parcel	your message, cut into packets (lesson 2)
The courier	the network and its **routers**
The address on the parcel	the **IP address**, written in each packet's header
Country, city, street (the building)	the **network part** of the address
The flat number	the **device part** (also called the host part)

Exactly where the line between the two parts falls is lesson 13, subnets and CIDR.

4 What an IP address looks like

Here is a real one: **192.168.1.5**. Four numbers, with dots between them. This kind is called **IPv4**, version four, and it's the one you'll see most. Each number goes from 0 to 255, because each one is stored in 8 bits, one byte. A **bit** is a single 0 or 1, and 8 bits can count 256 values: 0 (00000000) to 255 (11111111). So the whole address is four bytes: 32 bits. In this home network, 192.168.1 is the network part, the building, and 5 is this phone's flat number.



Each number is 8 bits = 1 byte, so it goes from 0 (00000000) to 255 (11111111): 256 values.

4 numbers × 8 bits = 32 bits. Where the line between the two parts falls is lesson 13 (subnets and CIDR).

Figure 2. Anatomy of an IPv4 address: four numbers, each one byte (0 to 255), 32 bits in all; here 192.168.1 is the network part and 5 the device part.

5 Real numbers: why we ran short

Question	Arithmetic	Answer
values in one number	2^8	**256** (0 to 255)
bits in an IPv4 address	4 × 8	**32 bits**
IPv4 addresses	256 × 256 × 256 × 256 = 2^{32}	**4,294,967,296** ≈ 4.3 billion
IPv6 addresses	2^{128}	**≈ 3.4 × 10^{38}** (a 39-digit number)

4.3 billion sounds like a lot, but there are over 8 billion people, many with a phone, a laptop and a TV, plus every server. The last big free blocks of IPv4 addresses ran out in 2011. There are two fixes. One: **IPv6**, with addresses 128 bits long, written in hexadecimal groups like `2001:0db8:85a3:0000:0000:8a2e:0370:7334`: enough for every grain of sand on Earth. Two: **private addresses**, reused in every home.

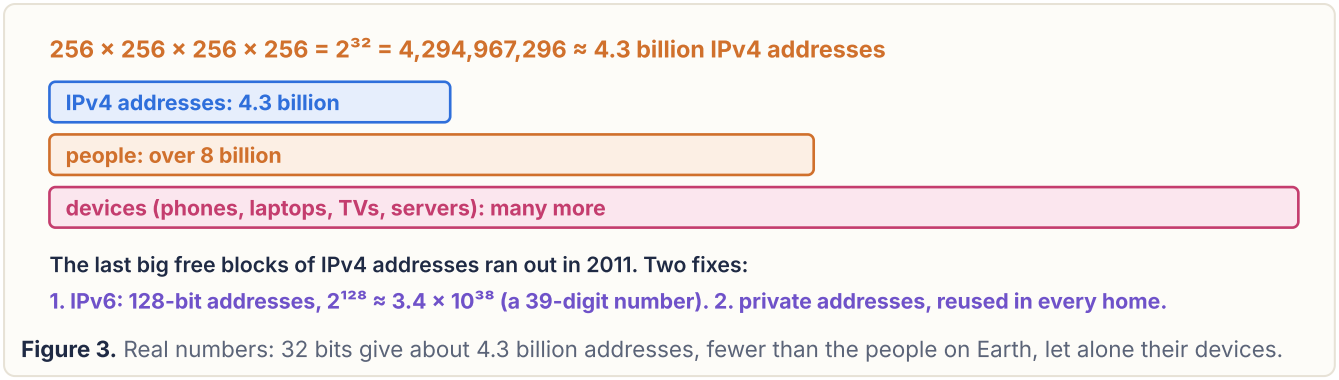


Figure 3. Real numbers: 32 bits give about 4.3 billion addresses, fewer than the people on Earth, let alone their devices.

6 At home: private addresses, one public address

Your Wi-Fi router gives each device a **private** address: your phone gets 192.168.1.5, your laptop 192.168.1.6 and the TV 192.168.1.7. Three ranges are kept for private use, and millions of homes reuse them; that's fine, because they never leave the home:

Private range	Written as (lesson 13)	Size
10.0.0.0 to 10.255.255.255	10.0.0.0/8	16,777,216 addresses
172.16.0.0 to 172.31.255.255	172.16.0.0/12	1,048,576 addresses
192.168.0.0 to 192.168.255.255	192.168.0.0/16	65,536 addresses

Your router also has one **public** address on its outside, for example 203.0.113.7. The whole internet can reach it, and to the internet your whole family is that one address. How the router shares it is lesson 14 (NAT, network address translation).

One special address: **127.0.0.1**, called **localhost**, always means this same device.

See your own: on a phone, open Wi-Fi settings and tap your network; on Windows, type `ipconfig`; on a Mac or Linux, `ifconfig` (or `ip addr` on Linux). That shows your private address. For your public address, search “what is my IP”.

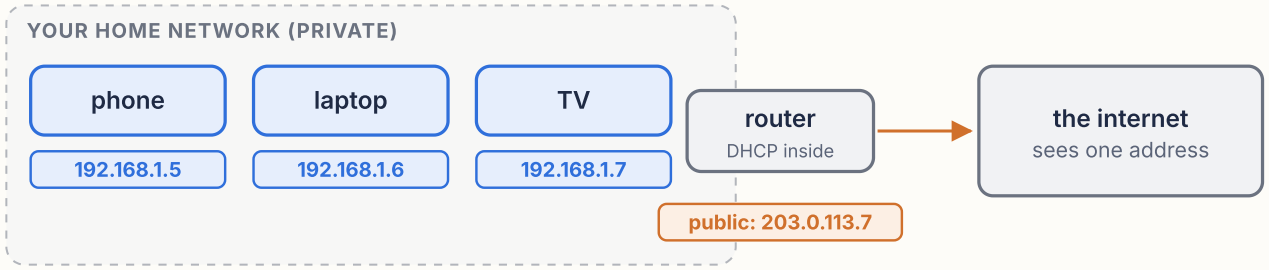
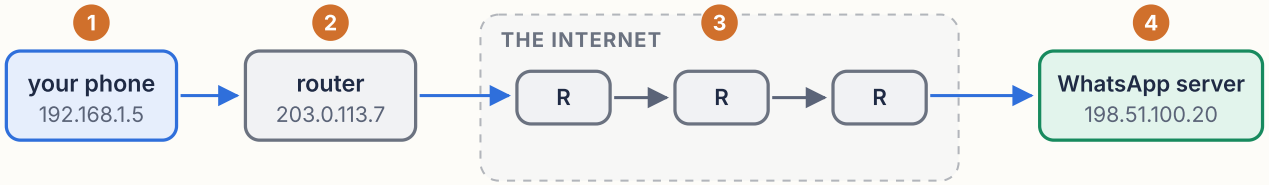


Figure 4. At home, the router gives each device a private address and shows the internet one public address. How that one address is shared (NAT) is lesson 14.

7 Step by step: one message

1. **Address it.** Your phone writes the header: from 192.168.1.5, to the WhatsApp server's public address. Job: say where to go, and where to reply.
2. **Leave home.** The router sees the destination isn't at home and sends the packet out, with the public address as the sender. Job: get the packet out of the building.
3. **Hop by hop.** Routers on the internet read the network part and pass the packet to one that's closer. Job: one step closer each time.
4. **Deliver.** The server's network uses the device part to reach that one server. The reply comes back the same way, to your phone.



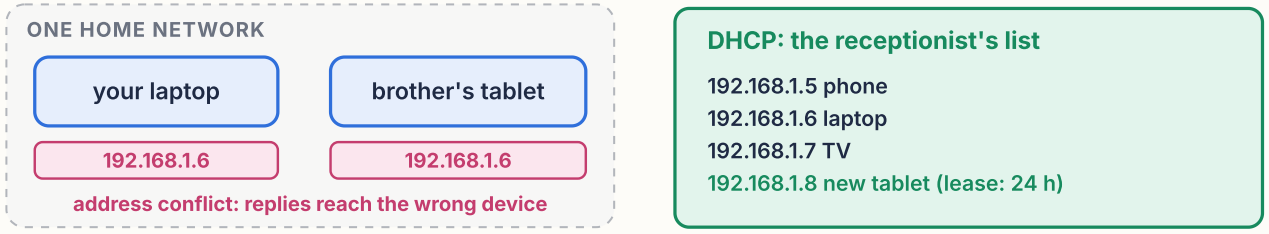
1 the phone writes from: 192.168.1.5, to: 198.51.100.20 2 the router sends it out, from: its public address
3 each router reads the network part and passes it one step closer 4 the device part finds the one server
The reply comes back to 203.0.113.7, and the router hands it to the phone.

Figure 5. One message, step by step: address it, leave home, hop by hop, deliver; the reply comes back the same way.

The addresses 203.0.113.7, 198.51.100.20 and 198.51.100.9 come from ranges reserved for examples in documentation (RFC 5737), so they belong to nobody real.

8 What goes wrong

- **An address conflict.** Your laptop and your brother's tablet both end up as 192.168.1.6: two doors with one flat number. The router can't tell them apart, replies go to the wrong device, and both connections keep breaking.
- **The fix: DHCP**, short for dynamic host configuration protocol. Think of it as the building's receptionist: a new device asks “which flat can I have?”, and DHCP picks a free number, writes it down, and never gives it to two devices at once. The number is lent, often for a day: a **lease**. At home, DHCP runs inside your router. Most conflicts come from someone typing an address in by hand.
- **A wrong address.** Send to an address where no device is, and there's no door to knock on. Nothing answers, and your app gives up with a timeout.



DHCP picks a free address, writes it down and lends it for a while (a lease), so it never gives one address to two devices.
Most conflicts come from an address typed in by hand. A wrong address, where no device is, gets no answer: a timeout.

Figure 6. What goes wrong: two devices with one address on one network. The fix is DHCP, which lends each device a free address and keeps a list.

PAUSE AND THINK

Your phone shows 192.168.1.5, and so does a friend's phone in another city. Is that a conflict? Cover the answer and decide.

Answer. No. Those are private addresses in two different homes: two different buildings. A conflict needs one address twice on the same network. On the internet, each home shows its own public address.



Same private address, two different networks: no conflict. A conflict needs one address twice on the same network.

Figure 7. The pause-and-think answer: two homes can both use 192.168.1.5; each shows the internet its own public address.

TWO COMMON MISTAKES

- **"Every device has its own address on the internet."** No. Most devices have a private address; the internet sees the router's one public address.
- **"A number in an address can be anything, like 300."** No. Each number is one byte, so it is always 0 to 255; 192.168.1.300 is not a valid address.

RECAP IN THREE LINES

1. An IP address names a device, with a network part and a device part.
2. IPv4 is four numbers from 0 to 255, 32 bits, about 4.3 billion addresses, so we also have IPv6 and private addresses.
3. At home, DHCP hands out unique private addresses, and the router shows one public address.

9 Interview questions: what to say in 30 seconds

Q1. What is an IP address, and what are its two parts? BASIC

An IP address identifies a device on a network, so packets can be delivered to it. It has a network part, which says which network the device is on, and a host (device) part, which says which device on that network. Routers use the network part to move a packet toward the right network; the host part finds the device at the end.

Q2. What does an IPv4 address look like, and how many are there? BASIC

Four numbers from 0 to 255 separated by dots, such as 192.168.1.5. Each number is 8 bits, one byte, so an address is 32 bits, and there are 2^{32} = 4,294,967,296 of them, about 4.3 billion. That is not enough for every device, which is why IPv6 (128 bits) and private addresses exist.

Q3. What is the difference between a public and a private address? BASIC

A public address is unique on the internet and reachable from it. Private addresses come from three reserved ranges, 10.0.0.0/8, 172.16.0.0/12 and 192.168.0.0/16; they are reused in millions of networks and are not routed on the internet. A home router gives its devices private addresses and shows the internet one public address.

Q4. Your laptop and another device on the same Wi-Fi have the same IP address. What happens, and how is it avoided? SCENARIO

An address conflict: the network cannot tell the two apart, replies reach the wrong device and both connections keep breaking. DHCP avoids it by handing out addresses from a pool, keeping a list and lending each one for a lease, so it never gives one address to two devices. Most conflicts come from an address typed in by hand.

Q5. Two phones in different cities both have 192.168.1.5. Is that a problem? TRICKY

No. Those are private addresses on two different networks. A conflict needs the same address twice on the same network. On the internet, each home shows its own public address.

Q6. What is 127.0.0.1? TRICKY

The loopback address, called localhost: it always means this same device. A program that sends to it is talking to itself, and the packets never leave the machine. (The whole 127.0.0.0/8 range is reserved for loopback.)

Read more. RFC 791 (Internet Protocol, version 4), RFC 1918 (private address ranges), RFC 2131 (DHCP), RFC 8200 (IPv6), RFC 5737 (example addresses). The full written lesson is at backendroadmap.com.

Next: SD-N-004 · Ports and sockets: which app gets the packet backendroadmap.com